

词汇: *biology* 生物学

- 1 Are you one of those people that sees faces in **inanimate** objects? Researchers at the National Institutes of Health have been trying to find out why this is. They have **pinpointed** a **circuit** in an **evolutionarily** ancient part of the brain. It's called the superior colliculus, and it helps us recognise facial features. It **triggers** movement of the eyes, turning of the head and enables us to **detect** faces.
- 2 This study helps us understand face pareidolia – the common psychological **phenomenon** where the brain **perceives illusions** of faces in objects, like seeing faces in the clouds, exteriors of houses or even in a slice of toast. A study carried out by Susan Wardle showed that the parts of the human brain that respond to faces are sensitive to the shapes of other objects, and that our visual system is **optimised** for detecting face-like structures.
- 3 To test whether the superior colliculus might help in face detection specifically, researchers **assembled** a collection of images, including faces, biological non-face objects, like hands and arms, and other **miscellaneous** items. They then showed these images to adult monkeys in their **peripheral field** and recorded **neural** responses in the superior colliculus. It was found that detection of faces was much faster and preferred by neurons. This explains how our brains do it.
- 4 But why do our brains do it? Well, it's mainly for survival. According to Dr Amanda Robinson at the University of Queensland, face pareidolia tells us a lot about how we recognise social **cues**. We can determine whether the person or thing looking at us is a **threat** because of their facial expression. Richard Krauzlis, senior author of the study, explained that "quick recognition is a key skill in humans and other **primates**", such as gorillas and chimps. Additionally, biologists have observed human facial recognition and neural developments among species ranging from dogs, sheep, birds and even some insects.
- 5 What we know about recognition of faces and expression can further inform research on conditions such as **autism**, where face detection and recognition are often **impaired** from early childhood. It also helps us understand why we think we saw a friendly face in the living room wallpaper.

测验与练习

一、将标题和段落配对。

Paragraph 1 _____

Paragraph 2 _____

Paragraph 3 _____

Paragraph 4 _____

Paragraph 5 _____

- a. How can we prove a preference for facial detection?
- b. What is face pareidolia?
- c. What more can we learn from facial detection?
- d. When did we discover facial detection?
- e. Where does facial detection happen?
- f. Why does the brain prefer facial detection?

二、根据文章内容选出正确的选项。

1. The circuit found in the brain has been there for a long time.

- a. True
- b. False
- c. Not given

2. Which of these functions is not carried out by the parts of the human brain that respond to faces?

- a. detecting facial features
- b. identifying parts of the body
- c. sensitivity to shapes

3. Researchers tested to see if monkeys' brains reacted to objects by...

- a. displaying large images of facial features, such as eyes, noses and mouths.
- b. hiding images of faces and having monkeys find them.
- c. showing them images of lots of different things.

4. What does Richard Krauzlis say about facial detection and recognition?

- a. It's a key skill shared by humans and primates.
- b. It's a skill that all animals can learn.
- c. It's a skill only humans possess.

5. The study can help other research on conditions for those who are _____.

- a. autistic
- b. expressive
- c. visually impaired

三、用方框中的单词完成对本文的总结。

The **1)** _____ connections in the brain can be described as a **2)** _____. When we see **3)** _____ objects, the shape of the object can **4)** _____ our brain into seeing facial features, creating a(an) **5)** _____ that isn't really there. This is called face pareidolia.

circuit	illusion	inanimate
neural		perceive
peripheral		trigger

（答案见词汇表后）

词汇表

inanimate	无生命的
pinpoint	准确指出
circuit	环路，线路
evolutionarily	进化上...的
trigger	触发
detect	识别
phenomenon	现象
perceive	注意到，察觉到
illusion	错觉
optimise	优化
assemble	汇集
miscellaneous	各式各样的，混杂的
peripheral field	外围视野
neural	神经系统的
cue	暗示，信号
threat	威胁
primate	灵长类动物
autism	自闭症
impaired	受损的

答案

一、将标题和段落配对。

Paragraph 1 **e) Where does facial detection happen?**

Paragraph 2 **b) What is face pareidolia?**

Paragraph 3 **a) How can we prove a preference for facial detection?**

Paragraph 4 **f) Why does the brain prefer facial detection?**

Paragraph 5 **c) What more can we learn from facial detection?**

二、根据文章内容选出正确的选项。

1. **a.** *They have pinpointed a circuit in an **evolutionarily ancient** part of the brain.*
2. **b.** *... the parts of the human brain that respond to faces are **sensitive to the shapes of other objects**, and that our visual system is optimised for **detecting face-like structures**. Nothing about identifying parts of the body is mentioned.*
3. **c.** *... researchers assembled **a collection of images, including faces, biological non-face objects, like hands and arms, and other miscellaneous items**.*
4. **a.** *Richard Krauzlis, senior author of the study, explained that "**quick recognition is a key skill in humans and other primates**", such as gorillas and chimps.*
5. **a.** *What we know about recognition of faces and expression could further inform research on conditions such as **autism**, where face detection and recognition are often impaired from early childhood.*

三、用方框中的单词完成对本文的总结。

The **neural** connections in the brain can be described as a **circuit**. When we see **inanimate** objects, the shape of the object can **trigger** our brain into seeing facial features, creating an **illusion** that isn't really there. This is called face pareidolia.